

The Backtest That Lied:

A Forensic Investigation of Leakage, Implementation Failure,
and Liquidity Dependence in A-Share Machine Learning

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1 Abstract

A near-perfect cross-sectional prediction signal—monthly rank IC 0.940 collapsed to 0.100 after four detected data leakages were corrected. The surviving signal, mean OOS IC 0.108 with Newey–West $t = 10.5$, represented genuine low-liquidity stock-selection ability. With executable portfolios and transaction costs, the strategy produced negative absolute returns in the liquid U50 universe (net annual -0.9%, max drawdown +61.3%). The OHLCV-only hypothesis was formally terminated. The primary contribution is methodological.

2 Introduction

A model can predict returns and still fail as a strategy.

The initial result: IC 0.940 across 96 consecutive OOS months collapsed by 89% after four leakages. Retained signal: mean OOS IC 0.108, Newey–West $t = 10.5$.

- (1) Cross-sectional predictability: IC 0.108 survived null infrastructure.
- (2) Signal concentrated in illiquid stocks: +48.1% of holdings in Q1. Within-Q1: +2.7% monthly excess.
- (3) Liquid universes unprofitable: U50 net -0.9%, max DD +61.3%.
- (4) Longer horizons failed: H3 -21.7%, H6 -15.2% net annual.

The OHLCV-only strategy was terminated. Offered as a research methodology case study.

3 Research Question

The primary research question is:

Can daily OHLCV data produce a statistically valid, economically executable, and scalable A-share trading strategy?

This question decomposes into three hierarchical tests:

T1: Is the predictive signal real? — Can cross-sectional return rankings be predicted out-of-sample with statistical significance, after correcting for known data leakages (survivorship bias, look-ahead bias, own-stock contamination, calendar encoding)?

T2: Can the signal survive execution and costs? — When translated into an executable portfolio with next-open timing, realistic buy/sell costs, and a 90/80 no-trade band, does the signal produce positive absolute returns?

T3: Can the signal survive liquidity and capacity constraints? — Does the strategy work in tradable universes (top 50% and top 30% by ADV20), or is performance dependent on stocks too illiquid to trade at meaningful scale?

Each test was evaluated against pre-registered gates. No gate was changed after seeing results. Experiments that failed a gate were classified as rejected; the hypothesis branch was terminated when all rescue attempts (residualization, multi-horizon targeting) failed the same gates.

4 Data and Methodology

4.1 Data

The research used the RQAlpha A-share daily market-data bundle, covering 5,511 historically listed stocks. A point-in-time (PIT) universe was constructed without survivorship filters, resulting in 4,073 stocks appearing across the full sample period. Daily OHLCV data was aggregated to monthly frequency.

Important: True point-in-time fundamental data (market capitalization, book value, earnings, industry classification) was unavailable. All analysis was conducted using price and volume data only. This limitation is fundamental to the project’s conclusion.

4.2 Feature Set

Twenty ranked technical indicators were computed at monthly frequency:

- **Momentum:** 5-, 10-, 20-, 60-, and 120-day returns (M5, M10, M20, M60, M120)
- **Volatility:** 5-, 20-, and 60-day standard deviation of returns (S5, S20, S60)
- **Volume:** 5- and 20-day average turnover ratios (V5, V20)
- **Moving Average:** 20- and 60-day MA deviation (MA20, MA60)
- **Channel Position:** Position within 20-day high-low range (POS)
- **RSI:** 14-day relative strength index (RSI)
- **Turnover:** Current and 5-day average turnover (TN, TN5)
- **Reversal:** 1- and 5-day reversal (REV1, REV5)
- **Cross-sectional:** 20-day cross-sectional rank (X20)
- **Skewness:** 20-day return skewness (SK20)
- **Breadth:** Number of advancing stocks over 20 days (N20)

All features underwent cross-sectional rank normalization to $[-1, 1]$ within each month. This normalization was recomputed separately for each validation month during training to prevent information leakage.

4.3 Model

LightGBM gradient-boosted trees were used with the following frozen hyperparameters:

Parameter	Value
Number of estimators	200
Learning rate	0.02
Maximum depth	3
Number of leaves	8
Subsample fraction	0.7
Column subsample	0.7
L1 regularization (α)	1.0
L2 regularization (λ)	10.0

A 20-seed ensemble (seeds 42–241) was used for each fold. No hyperparameter tuning was performed; the parameters were frozen before any results were observed.

4.4 Validation Protocol

Three temporally ordered walk-forward folds were used:

Fold	Training	Validation
Fold 1	2010–2016	2017–2019
Fold 2	2013–2019	2020–2021
Fold 3	2015–2021	2022–2024

4.5 Inference

- **Information Coefficient (IC):** Cross-sectional Spearman rank correlation between predicted and realized returns, computed monthly.
- **ICIR:** Mean IC divided by standard deviation of IC.
- **Newey–West:** Heteroskedasticity- and autocorrelation-consistent standard errors with 3 and 6 lags.
- **Block Bootstrap:** 1,000 resamples with block length of 6 months, 95% confidence intervals.

4.6 Null Infrastructure

Three adversarial tests were applied to every specification:

- (1) **iid Gaussian null:** Replace target returns with independent standard normal draws. Expected: $IC \approx 0$.
- (2) **Constant-label test:** Assign identical labels to all observations. Expected: standard deviation of predictions ≈ 0 .
- (3) **Permutation audit:** Shuffle labels across stocks within each month, retrain, check correlation with original predictions. Expected: ≈ 0 .

4.7 Portfolio Construction

Monthly rebalancing with 90/80 no-trade bands:

- At month-end: rank stocks by ensemble prediction.
- Select top 10% by rank percentile.
- Enter at ≥ 90 th percentile; retain at ≥ 80 th percentile.
- Execute at next trading-day open.
- Equal weight within portfolio.

4.8 Cost Scenarios

Scenario	Buy (bp)	Sell (bp)
Gross	0	0
Base	13	18
Stress	22.5	27.5

Cost components: commission 0.03%, stamp tax 0.05%, slippage 0.1%.

4.9 Evidence Architecture

All numerical claims in this report are generated from `data/evidence.json`, a machine-readable evidence layer containing 23 metrics, 3 collapse records, and full provenance information (experiment IDs, artifact paths, validation statuses). The `scripts/validate_evidence.py` validator and `tests/test_evidence.py` unit tests enforce structural integrity and reference resolution.

5 The Three Collapses

5.1 Collapse I: Statistical Illusion

Initial belief: 20 factors + LightGBM could predict returns with IC 0.940. Detection: Date-shift scan peaked at $k = 0$, not $k = +1$. Correction: IC collapsed $0.940 \rightarrow 0.100$. Interpretation: 89% of apparent predictive power was mechanical illusion (absolute decline: 0.84 IC points).

5.2 Collapse II: Economic Implementation

Initial belief: IC 0.108 should produce positive portfolio returns. Results: U50 executable (31bp costs): net -0.9%, DD +61.3%. Benchmark: -5.9%. Interpretation: Statistical predictability $\neq$ economic investability.

5.3 Collapse III: Scalability

Initial belief: Full universe profitable (U100 net +6.8%). Finding: +48.1% of holdings in Q1. Within-Q1: +2.7% monthly excess. Interpretation: The alpha existed where capacity was severely constrained.

6 Portfolio Implementation

Table 1: Portfolio performance under base cost scenario (31bp round-trip).

Portfolio	Net Annual	Sharpe	Max DD	Excess
midrule U100 RAW (base31)	+6.8%	+0.31	+46.3%	+6.9%
U50 RESID (base31)	-0.9%	-0.04	+61.3%	+4.9%
U50 EW benchmark	-5.9%	-0.26	—	—
bottomrule				

U50 RESID generates +4.9% annual excess over the benchmark (-5.9%). Positive excess over a poor benchmark is not skill.

Table 2: Liquidity quintile composition.

Quintile	Median ADV (M¥)	% Holdings	Net Annual
midrule Q1 (lowest)	19	+48.1%	+10.4%
Q2	41	25.8%	+3.9%
Q3	75	14.0%	-0.6%
Q4	145	7.8%	-3.5%
Q5 (highest)	386	4.2%	-2.6%
bottomrule			

7 Liquidity Attribution

Within-Q1 selection: +2.7% per month. The signal was concentrated in the lowest-liquidity stocks and was severely capacity-constrained. Maximum feasible capacity was not fully quantified.

8 Residual and Horizon Tests

8.1 Cross-Fitted Residual Target (RESID)

Own-stock contamination audit: IC collapsed from 0.760 to 0.120 after cross-fitting. Residualization improved U50 marginally (net -0.9%, DD +61.3%) but could not reach profitability.

8.2 Multi-Horizon Targets (H3 and H6)

H3 U50: -21.7% annual. H6 U50: -15.2% annual. Multi-horizon targeting does not rescue OHLCV-only scalability.

9 The Signal Was Real. The Strategy Was Not.

Table 3: Decision gates.

Criterion	Result	Gate
midrule Predictive ranking	Passed	IC > 0, NW $t > 2$
Time-aware significance	Passed	Bootstrap CI excludes 0
Within-Q1 selection	Passed	Excess > 0
Positive U50 return	Failed	Net < 0
Acceptable drawdown	Failed	DD > 25%
Scalable capacity	Failed	Q1 holds +48.1%
midrule Decision	TERMINATED	OHLCV branch closed
bottomrule		

Frozen baseline: IC 0.108, NW $t = 10.5$. Pre-registered gates prevented post-hoc optimization. Termination was the only honest decision.

10 Limitations

The following limitations apply to all claims in this report:

- (1) **Data scope:** Only daily OHLCV data was available. True point-in-time fundamental data (market capitalization, book value, earnings, accruals, industry classification, analyst

forecasts) was not accessible during the research period. The conclusion that OHLCV-only cannot produce a scalable strategy does not extend to strategies incorporating economically independent data sources.

- (2) **Unavailable market capitalization:** The ADV20-based liquidity findings should not be interpreted as size-effect findings. Without point-in-time market capitalization, it is not possible to distinguish between a liquidity effect and a true size effect.
- (3) **No industry classification:** Industry membership—a standard control in asset pricing tests—was unavailable. Industry-neutral rankings could not be tested.
- (4) **Non-redistributable data:** The RQAlpha market-data bundle is proprietary and not included in the project repository. This limits end-to-end reproducibility from raw data. The `data/evidence.json` file serves as the authoritative publication layer, providing all validated results with full provenance.
- (5) **Sample consumption:** The 2022–2024 period was used as the third validation fold. A genuinely unseen forward test would require data from July 2026 onward. The consumed sample cannot be reused for further hypothesis testing without inflating Type I error rates.
- (6) **Cost model:** Flat per-trade cost assumptions were used. In practice, costs vary with trade size, market conditions, and stock liquidity. The 31bp base scenario likely underestimates true costs in the lowest-liquidity quintile, meaning reported Q1 returns are upper bounds.
- (7) **Model family:** Only LightGBM was tested. Other model classes (neural networks, random forests, linear models with interaction terms) might extract different information from the same features. However, given the feature family’s fundamental limitation (OHLCV only), a model-class change was not expected to address the core problem.
- (8) **Factor breadth:** Only 20 technical factors were used. Factor discovery algorithms, alternative technical indicators, or order-book features were not explored. However, the project’s conclusion is about the information content of OHLCV data generally, not about a specific factor set.
- (9) **No capacity model:** A formal market-impact model was not constructed. Capacity estimates are based on ADV20 quintile boundaries and should be treated as approximate.
- (10) **Survivorship treatment:** The PIT universe included delisted stocks, but delisting returns were not explicitly modeled. Stocks that delisted during a holding period were treated as total losses, which is conservative.

11 Conclusion

OHLCV features contain genuine ranking information: IC 0.108, NW $t = 10.5$, within-Q1 selection +2.7% monthly excess. The signal did not survive executable portfolios: U50 net -0.9%, DD +61.3%. Multi-horizon: H3 -21.7%, H6 -15.2%. The contribution is a demonstration that clean negative results are more valuable than profitable-looking lies.

A Evidence Map

Key metrics (of 23 total):

- `target_leak_before_ic`: IC 0.940 → EXP_TARGET_SHIFT_01
- `target_leak_after_ic`: IC 0.100 → EXP_TARGET_SHIFT_01

- **frozen_baseline_mean_ic:** IC 0.108 $\rightarrow$ EXP_BASELINE_V43
- **u100_raw_net_annual:** +6.8% $\rightarrow$ EXP_DP1_PORTFOLIO
- **u50_resid_net_annual:** -0.9% $\rightarrow$ EXP_STEP5_ABLATION
- **q1_within_excess:** +2.7% $\rightarrow$ EXP_DP1A_QUINTILE
- **h3_u50_net_annual:** -21.7% $\rightarrow$ EXP_MH1R_HORIZON

B Experiment Ledger

1. **Target Timing:** IC 0.940 $\rightarrow$ 0.100 (89% collapse). Invalid $\rightarrow$ Validated.
2. **Cross-Fit Audit:** IC 0.760 $\rightarrow$ 0.120. Invalid $\rightarrow$ Validated.
3. **Frozen Baseline:** Mean OOS IC 0.108, NW $t = 10.5$. Validated.
4. **Null Infrastructure:** Permutation correlation ≈ 0 . Validated.
5. **Portfolio Diagnostics:** U100 +6.8%, U50 -0.9%. Q1-driven.
6. **Residual Ablation:** Cannot reach profitability. Rejected.
7. **Multi-Horizon:** H3 -21.7%, H6 -15.2% U50 net. Rejected.
8. **Termination:** OHLCV-only core strategy closed. Recorded.

C Reproducibility

C.1 What Is Reproducible

- **Evidence validation:** All metrics in this report can be validated by running `python scripts/validate_evi` and `python -m unittest discover -s tests` from the repository root.
- **Website rendering:** The interactive landing page (`index.html`) runs with `python scripts/serve.py` and loads all displayed values from `data/evidence.json`.
- **Report metrics:** All numerical claims in this report are generated from `data/evidence.json` by `report/build.py`. The generated macros in `report/../generated/` are the sole source of metric values.

C.2 What Is Not Fully Reproducible

- **Model training:** The original LightGBM training runs require the RQAlpha market-data bundle (~3.3GB), which is proprietary and not redistributable. Training scripts are included but data-dependent.
- **Raw data:** Daily OHLCV data is not included in the repository.
- **Prediction artifacts:** Per-stock, per-month prediction files are referenced by path in `evidence.json` but stored locally only.

C.3 Repository

<https://github.com/DresdenGman/the-backtest-that-lied>

Commit 70e7ec7 (v1.0.0) contains the frozen landing page and evidence package. This report corresponds to release v1.1.0.

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